

POWERFACTORY

PowerFactory 2024

Technical Reference

Thermal Ratings Definition

IntThrating

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

The thermal ratings object (*IntThrating*) is used to define for branch elements like lines, transformers, series reactors and capacitors, the nominal current or apparent power used to calculate the loading or in case when executing contingency analysis the short term loading.

The continuous ratings and the short term ratings can be defined depending on weather conditions like ambient temperature, solar irradiation (GHI) and dependent on the wind speed. For that a meteorological station (*ElmMeteostat*) must be selected for the corresponding line or transformer.

The meteorological station defines the weather conditions like the ambient temperature, wind speed and the solar irradiation (GHI). For these parameters also a corresponding time characteristic can be defined.

In case of no meteorological station is selected (or the selection is not supported), but the thermal ratings are weather dependent defined, the following assumption are made:

- Ambient temperature: $T_a = 20°$C
- Wind speed: $windspeed = 0$ m/s
- Global Horizontal Irradiance (GHI): $ghi = max(ghi)$ (highest defined ghi)

2 Configuration

The thermal ratings object can be configured at the Configuration page as follow:

- Unit
- Weather dependency
- Consider short term ratings

2.1 Unit

The unit for the entered value can be defined either in

- MVA: Apparent power in MVA
- kA: Current in kA
- %: percent of the corresponding nominal current of the element (line, series reactor, ...)

Note: The values are not converted to the new unit when switching the unit e.g. from % to MVA. The kA unit is not supported for 2-winding transformer models.

2.2 Weather Dependency

The continuous rating and the short term ratings can be defined depending on weather condition.

The following dependencies can be enabled or disabled:

- Ambient temperature in °C
- Solar irradiance (GHI) in W/m^2
- Wind speed in m/s

2.2.1 Ambient Temperature

The ambient temperature definition defines additional columns in the continuous and short term rating table.

The following figure shows an example for different ambient temperatures:

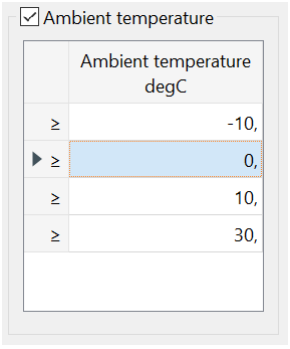


Figure 2.1: Configuration of ambient temperature

leads to $n+1$ column entries in the continuous and short term rating table (the addition column is here for temperature below -10 °C (first entry)).

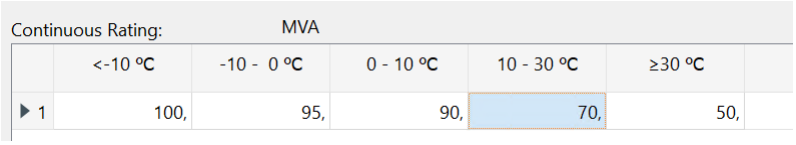


Figure 2.2: Continuous rating depending on ambient temperature

Note: The ambient temperature values must be entered in rising order.

2.2.2 Wind Speed

The wind speed definition defines additional rows in the continuous and short term rating table.

The following figure shows an example for the configuration for the wind speed:

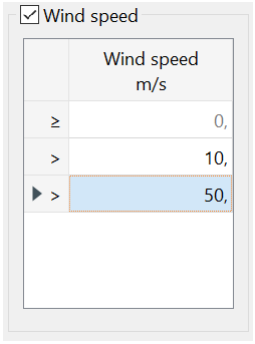


Figure 2.3: Configuration for wind speed

leads to n row entries in the continuous and short term rating table.

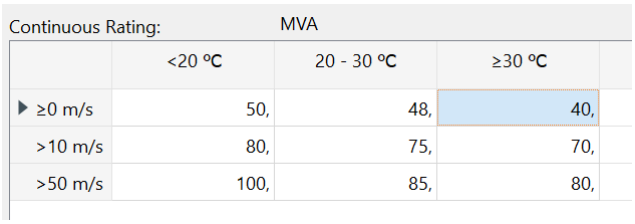


Figure 2.4: Continuous rating depending on wind speed and ambient temperature

If no ambient temperature dependency is configured, the table has just only one column.

Note: The wind speed values must be entered in rising order. The first entry is always 0 m/s and cannot be changed.

2.2.3 Solar Irradiance (GHI)

The wind speed definition defines additional rows in the continuous and short term rating table.

The following figure shows an example for the configuration for the solar irradiance (GHI):

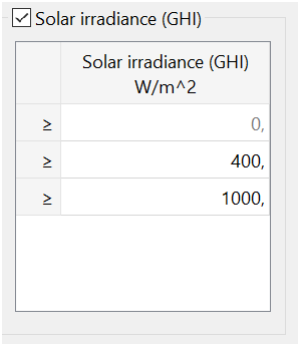


Figure 2.5: Configuration for solar irradiance (GHI)

leads to n row entries in the continuous and short term rating table.

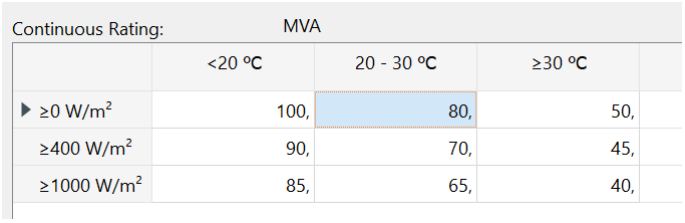


Figure 2.6: Continuous rating depending on solar irradiance (GHI) and ambient temperature

If no ambient temperature dependency is configured, the table has just only one column.

Note: The solar irradiance (GHI) values must be entered in rising order. The first entry is always 0 W/m^2 and cannot be changed.

2.3 Short Term Ratings

At the *Configuration* page, the definition of short term ratings can be enabled (Consider short term ratings). The short term ratings are depending on the pre-loading and duration (time phases).

2.3.1 Pre-loading and Duration

The pre-loading points are defined in % and the duration in minutes (min).

The following figure show an example of the configuration for the short term ratings:

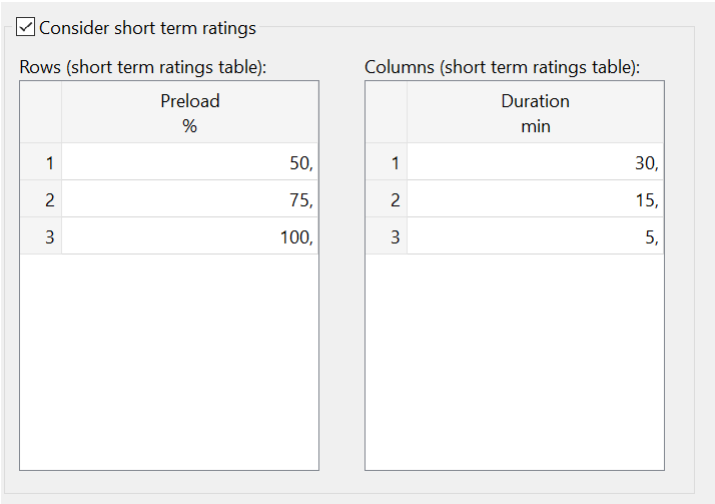
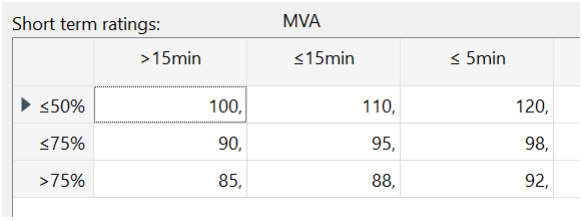


Figure 2.7: Configuration for short term ratings

leads to n rows (number of pre-loading points) and m columns (number of duration points):



Short term ratings:	MVA		
	>15min	≤15min	≤ 5min
≤50%	100,	110,	120,
≤75%	90,	95,	98,
>75%	85,	88,	92,

Figure 2.8: Short term ratings without weather dependency

Note:
- The Preload values must be entered in rising order. The value of the last point in the configuration table is replaced by $>$ of value of last point-1 (e.g. 100 % replaced by > 75 %).
- The Duration values must be entered in descending order. The value of the first point in the configuration table is replaced by $>$ of value of second duration (e.g. 30 min replaced by > 15 min).

When in one of weather dependency option the table will be resized as follow:

- Ambient temperature: additional columns are added
- Wind speed: additional rows are added
- Solar irradiance (GHI) additional rows are added

3 Continuous Ratings

If no weather dependency option is enabled, the parameter *ContRating* is used for the continuous rating.

3.1 Dimension/Layout

Dependent on the configurations the dimension of the continuous rating table (parameter *contRatMat*) is the following:

- Number of columns: $nTa + 1$ in case of ambient temperature dependency, where nTa is the number of ambient temperature points, otherwise 1.
- Number of rows:
  - 1 when no wind speed and solar irradiance (GHI) dependency.
  - $nSolar \cdot nWind$ when no solar irradiance (GHI) dependency then $nSolar = 1$ and when no wind speed dependency then $nWind = 1$.

The rows and columns are organized as follow:

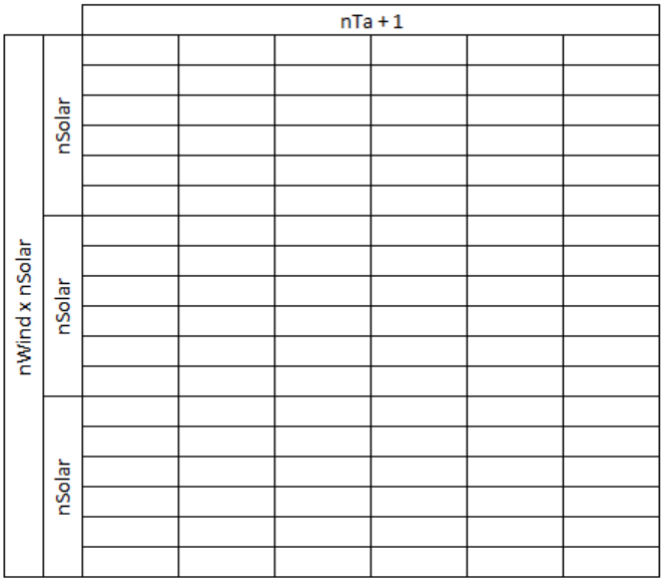


Figure 3.1: Dimension of the continuous rating table

The following figure shows an example with ambient temperature, wind speed and solar irradiance (GHI) dependency:

Continuous Rating: MVA

	<20 °C	20 - 30 °C	≥30 °C
▶ ≥0 m/s /≥0 W/m²	50,	48,	40,
≥0 m/s /≥400 W/m²	47,	45,	48,
≥0 m/s /≥1000 W/m²	45,	42,	35,
>10 m/s /≥0 W/m²	80,	75,	70,
>10 m/s /≥400 W/m²	77,	72,	63,
>10 m/s /≥1000 W/m²	72,	68,	60,
>50 m/s /≥0 W/m²	100,	85,	80,
>50 m/s /≥400 W/m²	96,	83,	78,
>50 m/s /≥1000 W/m²	93,	81,	73,

Figure 3.2: Example of continuous rating table

3.2 Weather dependent rating value

Depending on the settings of meteorological station (*ElmMeteostat*) the corresponding continuous rating value is determined to adapted the rated current of the corresponding element where the thermal ratings object is selected. The adapted rated current is then used to calculate the loading.

Note: Values are not interpolated, in borderline cases the more conservative cases are used.

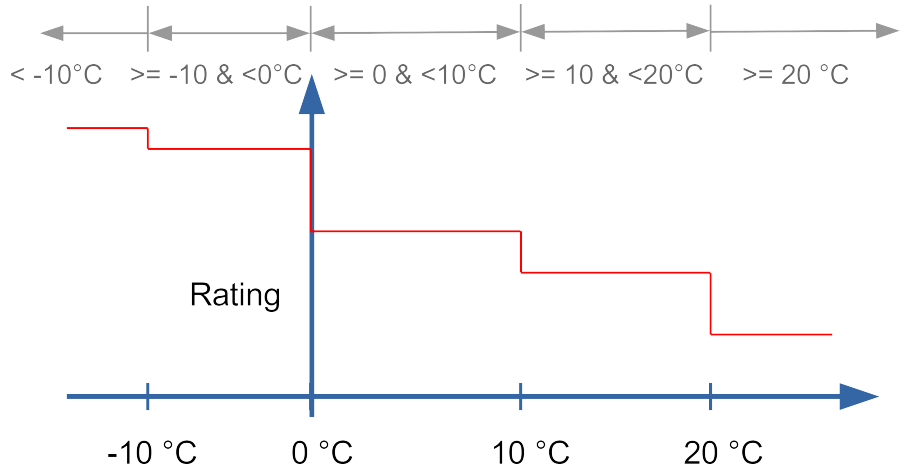


Figure 3.3: Example thermal ratings object dependent on ambient temperature

Note: The tolerance for the ambient temperature is 10^{-12} °C. If for example the actual temperature is $10 - 10^{-12}$ °C the ≥ 10 °C value is already used. In case of $10 - 2 \cdot 10^{-12}$ °C still the < 10 °C value is used. This leads to more conservative results in borderline cases.

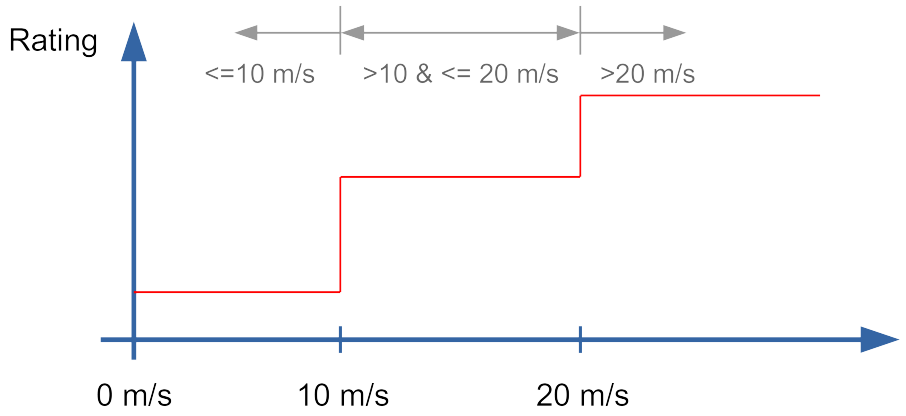


Figure 3.4: Example thermal ratings object dependent on wind speed

Note: The tolerance for the wind speed is 10^{-12} m/s. If for example the wind speed is $10+10^{-12}$ m/s the > 10 m/s value is used. In case of $10 + 10^{-13}$ m/s or exact 10 m/s still the ≤ 10 m/s value is used. This leads to more conservative results in borderline cases.

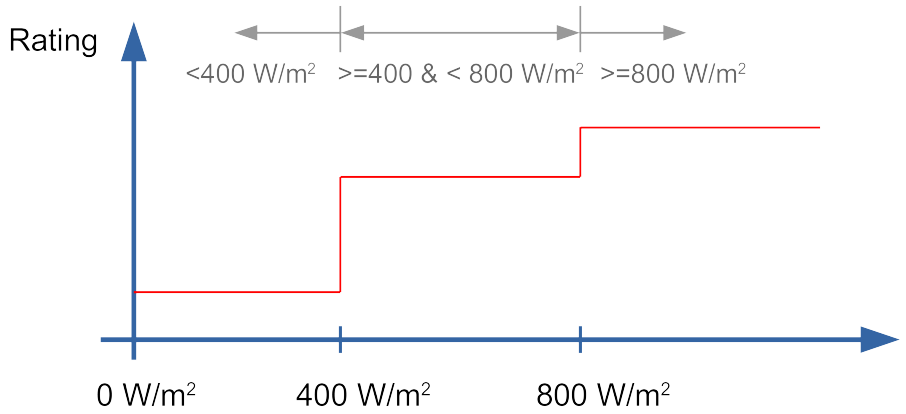


Figure 3.5: Example thermal ratings object dependent on solar irradiation (GHI)

Note: The tolerance for the solar irradiance (GHI) is 10^{-12} W/m². If for example the actual irradiance (GJI) is $ghi = 400 - 10^{-12}$ W/m² the ≥ 400 W/m² value is already used. In case of $ghi = 10 - 2 \cdot 10^{-12}$ W/m² still the < 400 W/m2 value is used. This leads to more conservative results in borderline cases.

3.3 Calculation of loading for branch elements

In principle, the loading of an element, is calculated for AC load flow calculation and RMS simulation by dividing the magnitude of the current through the nominal current.

$$loading = \frac{|I|}{I_{nom}} \cdot 100 \quad \% \tag{1}$$

The nominal current is based on the rated current of an element and in case of selected thermal rating object adapted.

- if the rating is entered in MVA:
  - 3-phase AC:

$$I_{nom(bus(x))} = S_{thrat} / \left(\sqrt{3} \cdot U_{n(bus(x))} \right) \cdot n_{parallel} \quad \text{kA}$$

  - 2-phase AC:

$$I_{nom(bus(x))} = S_{thrat} / U_{n(bus(x))} \cdot n_{parallel} \quad \text{kA}$$

  - 1-phase AC:

$$I_{nom(bus(x))} = S_{thrat} \cdot \sqrt{3} / U_{n(bus(x))} \cdot n_{parallel} \quad \text{kA}$$

  - 1-phase connected on a AC/BI system:

$$I_{nom(bus(x))} = S_{thrat} \cdot 2 / U_{n(bus(x))} \cdot n_{parallel} \quad \text{kA}$$

- if the rating is entered in kA:

$$I_{nom(bus(x))} = I_{thrat} \cdot n_{parallel} \quad \text{kA}$$

- if the rating is entered in %:

$$I_{nom(bus(x))} = I_r \cdot \%_{thrat}/100 \quad \text{kA}$$

where:

- S_{thrat} is the rated power determined by the thermal ratings object if the unit is configured in MVA.
- I_{thrat} is the nominal current determined by the thermal ratings object if the unit is configured in kA.
- $\%_{thrat}$ is the percentage value determined by the thermal ratings object if the unit is configured in %.
- $n_{parallel}$ is the number of parallel lines, transformers, ... (see *Basic Data* page of the corresponding element)
- I_r is the rated current of the corresponding element (contains already the number of parallel line, transformers, ...). See corresponding technical references. of the elements.
- $bus(x)$ is the connection side e.g. $busi$, $busj$ for lines or $bushv$, $buslv$ for transformers ...
- $U_{n(bus(x))}$ is the nominal voltage of the corresponding connected busbar of side $bus(x)$ in kV.

Note: The nominal currents for lines, series reactors/capacitors, switches are generally the same ($I_{nom(busi)} = I_{nom(busj)}$), but not if the line/cable is connected between different voltage levels ($U_{n(busi)} \neq U_{n(busi)}$).

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4 Short Term Ratings

4.1 Dimension/Layout

Dependent on the configurations the dimension of the short term rating table (parameter: *MyMatrix*) is the following:

- Number of columns: $(nTa + 1) \cdot nDuration$ in case of ambient temperature dependency, where nTa is the number of ambient temperature points and $nDuration$ is the number of duration points. If the ambient temperature dependency is disabled then $nTa = 0$.
- Number of rows:
  - Number of pre-loading points ($nPreload$) if no wind speed and mo solar irradiance (GHI) dependency.
  - $nPreload \cdot nWind \cdot nSolar$ when no solar irradiance (GHI) dependency then $nSolar = 1$ and when no wind speed dependency then $nWind = 1$.

The rows and columns are organized as follow:

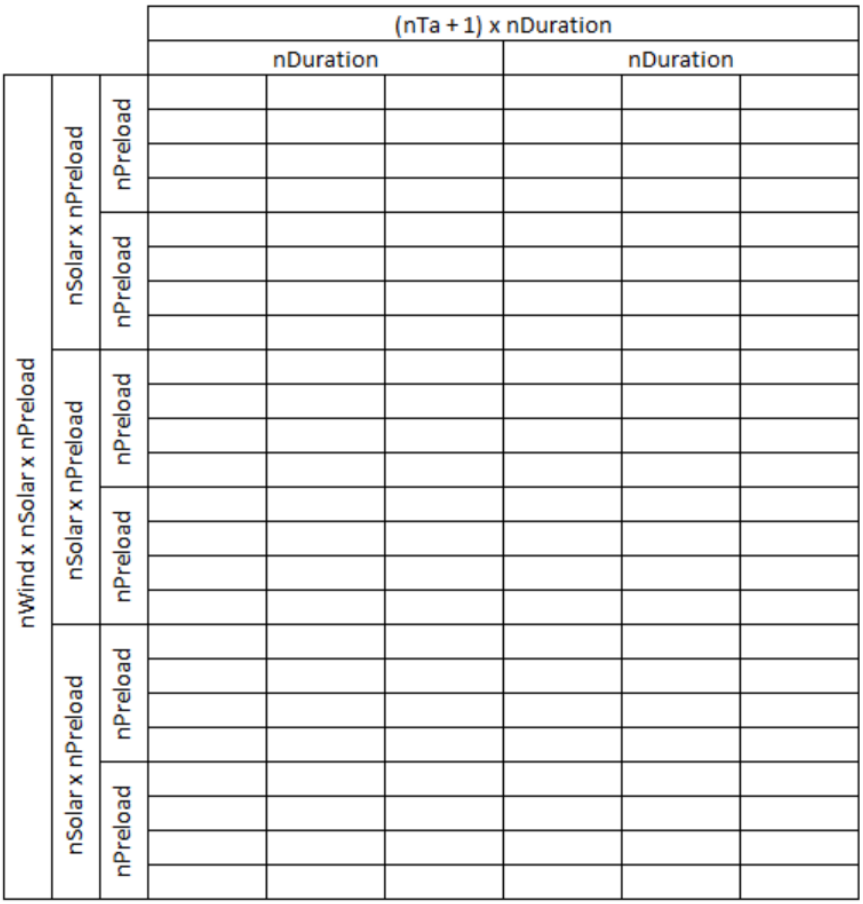


Figure 4.1: Dimension of the short term rating table

The following figure shows an example with ambient temperature, wind speed and solar irradiance (GHI) dependency:

Short term ratings: MVA

	<20 °C />15min	<20 °C /≤15min	<20 °C /≤ 5min	≥20 °C />15min	≥20 °C /≤15min
▶ ≥0 m/s /≥0 W/m² /≤50%	100,	120,	150,	95,	115,
≥0 m/s /≥0 W/m² /≤75%	90,	110,	140,	85,	105,
≥0 m/s /≥0 W/m² />75%	80,	100,	130,	75,	95,
≥0 m/s /≥1000 W/m² /≤50%	95,	115,	145,	90,	110,
≥0 m/s /≥1000 W/m² /≤75%	85,	105,	135,	80,	100,
≥0 m/s /≥1000 W/m² />75%	75,	95,	125,	70,	90,
>50 m/s /≥0 W/m² /≤50%	97,	113,	143,	92,	107,
>50 m/s /≥0 W/m² /≤75%	87,	103,	133,	82,	97,
>50 m/s /≥0 W/m² />75%	77,	97,	123,	72,	67,
>50 m/s /≥1000 W/m² /≤50%	95,	115,	145,	90,	110,
>50 m/s /≥1000 W/m² /≤75%	85,	105,	135,	80,	100,
>50 m/s /≥1000 W/m² />75%	75,	95,	125,	70,	90,

Figure 4.2: Example of short term rating table

Note: The column for ≥ 20 °C/≤ 5 min is not displayed on that figure.

4.2 Weather, Preload, Duration dependent rating value

Depending on the base case loading (preloading) and the time phase of the contingency, the corresponding short term rating value is determined.

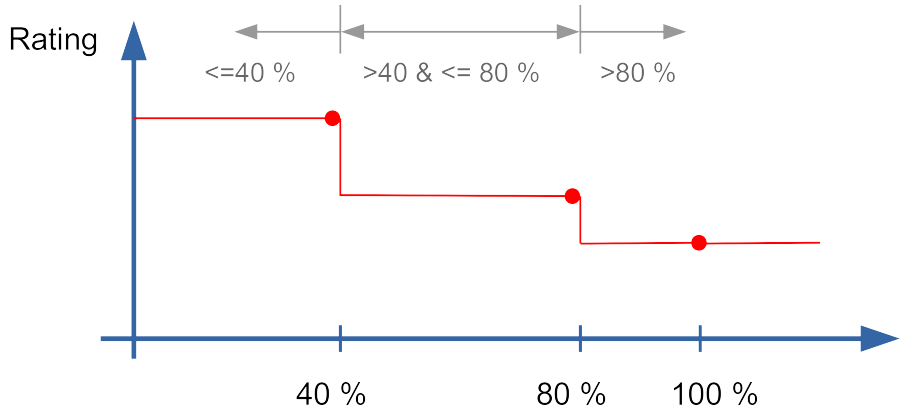


Figure 4.3: Example thermal ratings object dependent on preload in %

Note: The tolerance for the preload is 10^{-4} %.
If for example the actual preload is $preload = 40.0001$ % the ≤ 40 % value is still use. In case of $preload = 40.0002$ % the > 40 % value is used.

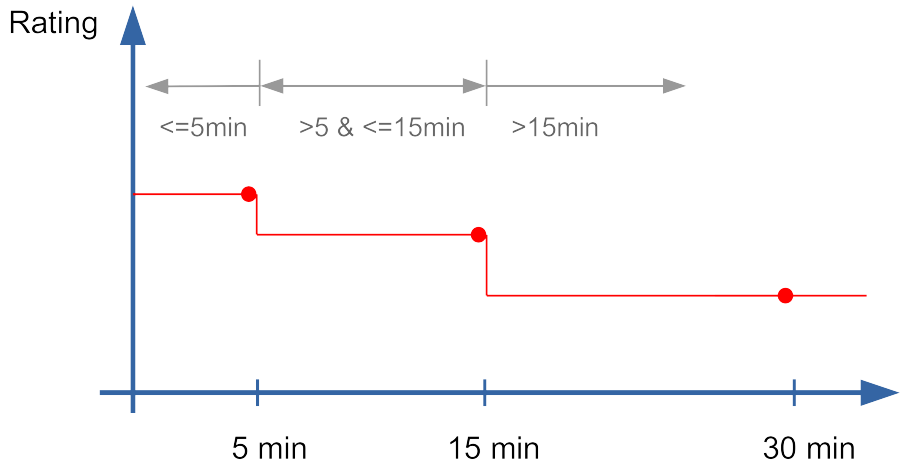


Figure 4.4: Example thermal ratings object dependent on duration (time phases) in min

Note: The tolerance for the duration is 10^{-15} min.
If, for example, the duration is $duration = 15 + 10^{-15}$ min the ≤ 15 min value is still used. In case of $duration = 15 + 2 \cdot 10^{-15}$ min the > 15 min is used.

For weather dependency, see Section 3.2.

4.3 Calculation of short term loading for branch elements

The short term loading ($loading_st$) is calculated like the loading (see Section 3.3), but the magnitude of the current is the post-contingency current used and the base case pre-loading and contingency time phase criteria is also used to determine the corresponding rated value of the thermal ratings object.